Outline of a Decision Support System Using Text Mining Technology

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Abstract

Recently, the acquisition and analysis of text have attracted much attention. How to obtain and analyze the large and varied text sources with reputation information, for instance questionnaires, Internet bulletin boards and call center consumer opinions, is an important question. The related works show that most research only classifies the opinions into bipolar orientation. In contrast to these, this study proposes a decision support system using Text Mining technology to extract and analyze the reputation information automatically from blog. This system first obtained the blog content by information retrieval, and saved the data in a database after Blog analysis and Blog classification. Then suitable products were recommended based on the user's selection and the database. In this paper, an evaluation of this system is also described.

1. Introduction

With the rapid growth of the Internet, personal opinions regarding products (cameras, cars, books, movies, electronics, etc.) are spread very quickly on the Web. Obviously, there is a lot of valuable information in these opinions. Effective use of this information would be beneficial to both consumers and manufacturers. Consumers would know which product is suitable and manufacturers would know their competitors' performances.

Blog, as one such information source has attracted much attention for these reasons. As we all know, the personal opinions in blog, such as feedback surveys, bulletin boards, opinions and product reviews, make up an enormous amount of unstructured and/or semi-structured data. This information is factual, but in free format and mostly textual, which is very difficult to manage in a short time. Therefore, there is a great need for some techniques to handle such unstructured data effectively. Text mining, especially opinion mining, can play an important role in fulfilling this need [1, 7].

Related works have mostly studied how to classify the opinions into bipolar orientation (i.e. either positive or negative opinion) [1, 2, 4, 6]. These research results make it possible for users to reduce the time spent on reading threads of text and focusing more on analyzing summarized information. On the other hand, the classification results could not offer a comparison of the same kind of product and also could not recommend a suitable product according to the requirements of consumers.

To address the above question, this study proposes a decision support system based on the product opinion in blog. The target language of this system is Japanese. Consumers can get a product recommendation directly through this system and can thus save a lot of time and costs. Instead of classifying the whole review, we analyzed the opinions on the phrase level. Another point in this study is the classification approach. It is necessary to classify the contents of blog into feature types(i.e. price, design or function) and P/N (positive or negative) orientation. We implemented feature classification using a supervised approach (machine learning) and P/N Classification using an unsupervised approach (semantic orientation). Moreover, we integrated Information Retrieval, DB and NLP techniques to realize this system. Although the precision is still poor, this system has a entire structure, from information collection, through information analysis to product recommendation. We will continue to improve it in the future.

The remainder of this paper is organized as follows: Section 2 provides the background of this study. Section 3 gives an overview of the proposed system, and describes the system in detail. Section 4 discusses the experimental evaluation of this system. Section 5 presents our conclusions.

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2. Background

In this section, we review the mining technology, text classification approaches, decision support system and related works.

2.1 Mining Technology

Text mining is a cross between Information Retrieval (IR), Data Mining (DM), Artificial Intelligence (AI), and Natural Language Processing (NLP). It is also known as intelligent text analysis, text data mining or Knowledge Discovery in Text (KDT), and generally refers to the process of extracting information and knowledge, facts and meaningful relationships, from text documents. Since most company data (over 80%) is stored as text, text mining is believed to have a high commercial potential value [9].

Opinion mining is a subdiscipline of text mining. The purpose of opinion mining is to mine reviews of various products by classifying them into positive or negative meaning from the 'subjective' terms contained in the document. Automated opinion mining can provide a quick search and analysis result to both consumers and manufacturers [1].

Web mining is the application of text mining techniques to extract the valuable knowledge from varieties of data on the Web, including logs, cookies, links, and contents. The current research area of Web mining includes 1) Web content mining, 2) Web structure mining, 3) Web usage mining [8]. In this study, Web content mining, or more specifically the mining of blog, is studied.

2.2 Text Classification

Text classification is one of the core problems in Text Mining [9]. It is a technique for dividing texts into two or more categories. For example, it is used for automatically assigning news articles to categories, such as politics, economy, and sport, based on the contents. There are many applications for classifying texts, for instance, analyzing the results of questionnaires, product opinions on Internet bulletin boards, or classifying e-mail as spam mail or not.

Text classification methods can be divided into two types: supervised classification approaches and unsupervised classification approaches. Supervised classification is a machine

learning technique for creating a function from training data. The training data consist of pairs of input objects and desired outputs. There exist many supervised learning methods such as Support Vector Machines (SVM), Naive Bayes classifiers etc. Unsupervised classification on the other hand does not require a prior training corpus. For example, the semantic orientation approach measures how far a word is inclined towards positive and negative to classify a review as recommended or not recommended [1].

2.3 Decision Support System

Decision Support System (DSS) is a specific class of computerized information system that supports business and organizational decision-making activities. A properly designed DSS is an interactive software-based system intended to help decision makers compile useful information from raw data, documents, personal knowledge, and/or business models to identify and solve problems and make decisions. DSS is also called a Knowledge-based system or an expert system. It analyzes data using symbolic logic, has an explicit knowledge base, and has the ability to explain conclusions in a way that users can understand. DSS can be useful to remind an experienced decision maker of options or issues to consider and to help a user make a complex decision [10].

2.4 Related Works

The problem of "opinion mining" has seen increasing attention over the last three years, from [1, 2, 4] and many others.

Turney [2] works on opinion classification using word semantic orientation. He presents an unsupervised learning algorithm called SO-PMI (Semantic Orientation Using Pointwise Mutual Information) for classifying reviews as recommended or not recommended. The classification of a review is based on the average semantic orientation of the phrases in the review that contain adjectives or adverbs. Kobayashi et al. [4] define the main portions of an opinion as <object, attribute, opinion>. They propose a technique for efficiently building an object name dictionary, an attribute expression dictionary, and an opinion word dictionary for any object domain. Chaovalit's study [1] gives a comparison between a supervised approach (machine learning) and an unsupervised approach (semantic orientation) on movie opinion mining.

3. Proposed System

In this section, we first give an overview of the proposed system and then describe the details of the process.

3.1 System Outline

The proposed system aims to collect and analyze opinions on a specific type of product automatically from blog, and then compare different products of this type, finally recommend a suitable product through interaction with the user.

An overview of this system is shown in Figure 1. This system is divided into two parts. Part one is information collection including step one to three. These steps will be repeated until the necessary information has been collected. Part two gives a recommendation to the user in accordance with the database information.

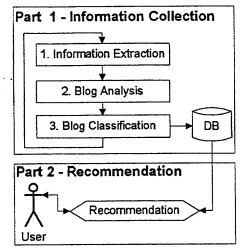


Figure 1: Flow of the system

3.2 Information Extraction

The first step is Information Search. We used the Google Search Engine to get all the information on one type of product in the blog on the Internet. The search keyword is the product's type and maker. The search URL's range is restricted by the URL's type(e.g. blog, bbs, review). The Lynx, a text browser was used to get the contents of the pages returned by the search engine.

These pages also usually contain other contents than the information about the product in question. Thus, it is necessary to extract the interesting blog information from the rest of the contents. Through analyzing various pages, we found that online sales sites have many replies about one topic, and personal blog pages are mostly the opinions of author themself. Therefore, to extract the required information efficiently, we used the Blog Contents Extraction Method presented in Figure 2. If the URL type is "online sales site", the review would be extracted by that page's style (e.g special keyword or title). On the other hand, if the URL type is "personal Blog site", the parts of the blog that contains the product information would be extracted using the date when the blog was edited.

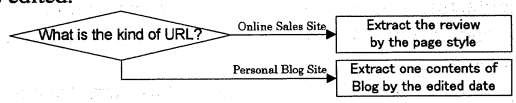


Figure 2: Blog Contents Extraction Method

3.3 Blog Analysis

Blog analysis is used to extract the opinion phrases through morphological analysis and phrase analysis. There are four steps in this process. The first is Sentence Separation. All the sentences were separated by end punctuation (such as "!" "." "。" ""?" etc). The second step is morphological analysis. We used a famous parser for Japanese called Chasen [11]. Furthermore, we added an Unknown Corpus to Chasen to increase the precision of the analysis. Most unknown words are product types or names of product makers. The third step is preprocessing of opinion collection. It means filtering stop words (such as "Person", "Year" etc), and merging the words that have the same POS (a part of speech). Finally, based on the grammatical structure of Japanese and the results in related works [4], an opinion generally consists of the object's name and opinion phrases. And the object's name is generally noun. The opinion phrase is generally adj / verb / noun. Therefore, we designed the following templates to collect the opinion phrases.

< noun + auxiliary word + adj / verb / noun >
< adj + noun / undefined / verb >
< noun + verb >
< noun + symbol + adj / verb / noun >
Except the above < adj >

3.4 Blog Classification

After extracting the opinion phrases, the classification of every phrase is necessary. The extracted phrases from Blog Analysis are classified and saved in the database. Two kinds of classification are done.

3.4.1. Feature Classification

Feature classification means to classify the phrase into Price(F1), Design(F2) or Function(F3) features. The Naive Bayes Machine Learning approach was used for classification. The Naive Bayes classifier, as a simple probabilistic classifier, is among the most successful known algorithms for learning to classify text documents. Although the Support Vector Machines (SVM) algorithm has higher-precision than Naive Bayes, the training time is long. From the comparative experiment in the next section, we can confirm that they had similar precision.

3.4.2. P/N Classification

P/N Classification aims to classify the phrases into positive or negative meanings using the SO (Semantic Orientation) value. We used and improved the SO-PMI (Semantic Orientation Using Pointwise Mutual Information) approach to perform this classification.

The SO-PMI algorithm is employed to estimate the semantic orientation of a phrase by measuring the similarity of pairs of words or phrases [2]. The Pointwise Mutual Information (PMI) between two words, $word_1$ and $word_2$ is defined as follows:

$$PMI(word_1, word_2) = \log_2\left[\frac{p(word_1 \,\&\, word_2)}{p(word_1)p(word_2)}\right] \quad (1)$$

SO of a phrase is calculated here as follows:

$$SO(phrase) = PMI(phrase, \text{"excellent"}) - PMI(phrase, \text{"poor"}) \quad (2)$$

The reference words "excellent" and "poor" were chosen because, SO is positive when *phrase* is more strongly associated with "excellent" and negative when *phrase* is more strongly associated with "poor".

Let hits(query) be the number of hits returned, the following estimate of SO can be derived from equations (1) and (2) with some minor algebraic manipulation. The Google Search Engine was used to get the hits counts.

$$SO(phrase) = \log_2\left[\frac{hits(phrase \quad NEAR \text{ "excellent"}) * hits(\text{"poor"})}{hits(phrase \quad NEAR \text{ "poor"}) * hits(\text{"excellent"})}\right] \quad (3)$$

The SO of a given phrase is calculated by comparing its similarity to a positive reference word ("excellent") with its similarity to a negative reference word ("poor") according to Formula 3. The SO of a phrase is positive if it is associated more strongly with "excellent" than "poor" and negative if it is associated more strongly with "poor" than "excellent". The formula corresponding to Japanese is shown in **4.3.2**. The SO values of all phrases are stored in a database.

3.5 Database Design

For later use in giving recommendations to users, the opinion information of products is stored in a database after extraction and classification. The database schema is shown in Figure 3.

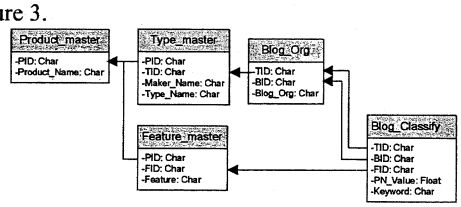


Figure 3: Database Schema

- [Product_master] Table:
  - PID: The identification of the product
  - Product_name: The name of the product
- [Type_master] Table:
  - PID: The identification of the product
  - TID: The identification of the product's type
  - Maker_name: The name of the maker
  - Type_name: The name of the type
- [Feature_master] Table:
  - PID: The identification of the product
  - FID: The identification of feature
  - Feature: The Blog Feature
- [Blog_Org] Table:
  - TID: The identification of the product's type
  - BID: The identification of the blog Sentence
  - Blog_Org: The contents of one blog Sentence
- [Blog_Classify] Table:
  - TID: The identification of the product's type
  - BID: The identification of the blog Sentence
  - FID: The feature of the keyword
  - PN_Value: The P/N (SO) Value of the keyword
  - Keyword: The keyword of the blog Sentence

3.6 Recommendation

Users are given recommendations based on the database and dialogue templates. Firstly, the system displays a product menu, where the user can select what he wants to investigate. Next, the user selects the feature (such as, All, Price, Design, Function) with which he is concerned. Through this interaction with the user, the system can understand the user's needs. Then, a suitable product is calculated using SQL statements accessing the database. This product is then recommended to the user. Finally, as a motivation of the recommendation, the blog sentence which has the best P/N value is presented. An example of this process is showed in Figure 4.

```
[S:] 下記の商品についてBlog情報を集めています。調べたい商品の番号を入力してください。
     (1) 電子辞書      (2) MP3      (3) Windowsノート
[U:] 2
[S:] MP3についてどんな特徴が気になりますか。
     (0) 全般      (1) 価格      (2) デザイン      (3) 機能
[U:] 2
[S:] デザインをお考えでしたら、下記の商品をお勧めします。
     APPLE iPod MA004J/A
[U:] ?
[S:] 下記のような評判が多いからです。
★: 軽いし、薄いし、いっぱい曲を持ち運べるipod nano・・・気に入ってます
```

Figure 4: A Recommendation to a user

4. Experimental Evaluation

4.1. Information Extraction

Table 1 shows the five types of IC Dictionary that were used in the prime experiments. Extracting the opinion information from the same Hit URLs manually took approximately 20 times as long as the system transaction time. Compared to the manual work, the system had a coverage of 87% on the extracted contents.

Table 1 : Information Extraction for five types of IC Dictionary

Type (IC Dictionary)	Hit URLs	Extracted Content (bytes)	Transaction time (min)	Manual time (min)
T1: SONY EBR-S7MS	267	18801	6	76
T2: CASIO XD-ST6200	48	13277	2	36
T3: SEIKO SR-E8500	169	16024	3	88
T4: SEIKO SR-E9000	254	18152	4	105
T5: SHARP Papyrus PW-A8400	753	29358	8	153
Average	298	19122	4.6	91.6

4.2. Blog Analysis

Here we evaluated analysis performance from the in-

formation retrieval perspective. Recall was measured as the ratio between the number of phrases correctly extracted as opinion and the total number of opinion phrases in the selected blog. Precision was measured as the ratio between the number of phrases correctly extracted as opinion and the total number of extracted phrases. The obtained Recall was 71% and the Precision was 52%. The Precision was poor because general words were also extracted as opinion.

4.3. Blog Classification

4.3.1. Feature Classification

As for feature classification, we compared the Naive Bayes algorithm and the SVM algorithm by classifying the same phrase into Price(F1), Design(F2) or Function(F3). Table 2 shows the comparative results. Since the Precision of the two algorithms is very close, and the speed of the application is important in this system, we chose Naive Bayes as the Feature Classification Method.

Table 2: A comparison of Navie Bayes and SVM

Average(F1,F2,F3)	Precision	Recall	Training Time	Execution Time
Naive Bayes	43.32%	71.85%	2 sec	1.5 sec
SVM	41.75%	72.38%	3.5 sec	2.5 sec

As we all know, Both Naive Bayes and SVM are a Machine Learning approaches. So it is necessary to prepare a training corpus in advance. We created the corpus by using the representative phrases in the blog about Price(F1), Design(F2) and Function(F3). These phrases were labeled manually in order to get an ideal training corpus. In this corpus, the number of Price(F1) representative phrases is 30, the number of Design(F2) representative phrases is 70, and the number of Function(F3) representative phrases is 210.

4.3.2. P/N Classification

As we mentioned in **3.4.2**, the SO-PMI approach was used in P/N Classification. Following Turney's study and the characteristics of the Japanese Language, we improved the SO formula to the one shown in Formula 4.

$$SO(phrase) = \log_2\left[\frac{hits(phrase\ NEAR\ good_basic) * hits(bad_basic)}{hits(phrase\ NEAR\ bad_basic) * hits("good_basic")}\right] + threshold \quad (4)$$

The reference words "excellent" and "poor" were replaced with "good_basic" and "bad_basic". The "good_basic" means the basic popular words for positive opinion in Japanese. In contrast, the "bad_basic" are the basic popular words for negative opinions in Japanese. The following words were used.

"good_basic" = "よい|いい|良い|満足|良く|好き|優れて|お薦め|利点|悪くない|いいかも|嫌いじゃない|傑作|大好き|良かった |満たしている"

"bad_basic" = "悪い|嫌|不満|嫌い|不安|不良|最悪 |悪く|欠点|満足ではない|好きではない|不具合|欠陥|悪質|不快|気に入らない|よくない|認めない|許せない|あまり|良くない| 満たしていない"

In a preliminary experiment, we used zero as a threshold to separate positive and negative meanings. The result of this trial on a small sample data was extremely unsatisfying. Both positive and negative phrases seem to lean towards the negative end. Possibly, in the time since Turney's study factors in calculating the SO have changed. It is also possible that different information sources give different results.

Therefore, we decided to setup our own baseline to separate positive SO from negative SO. The threshold was adjusted until a better result was obtained. The threshold used was the following:

$$threshold = \log_2\left[\frac{hits(bad_basic)}{hits(good_basic)}\right] * (-0.6) \quad (5)$$

After this improvement, we achieved the better precision, which is shown in Table 3. The precision of positive opinions was 70.91%, which is quite good. The precision of negative opinions was only 33.75%. In the Blog analysis process, general words were extracted as opinion phrases as mentioned in **4.2**. We believe this is one of the causes that negatively influenced the precision of the classification. Another cause is that the analysis of Japanese negative expressions is still insufficient.

Table 3: Precision of P/N Classification

	Positive (P) Precision	Negative (N) Precision	Average
Preliminary experiment	21.02%	21.69%	21.35%
After improvement	70.91%	33.75%	52.33%

4.4. Recommendation

For evaluating the overall performance of this system, we collected another 10 product examples, from the two categories MP3 and Windows NotePC. As the baseline of evaluation, we also gathered the User Review Graphs for the same products from http://www.kakaku.com/ on 2006/04/28. The review about price was judged from the price range in kakaku.com. These Graphs and products list are shown in Appendix.

Using the recommendation procedure described in **3.6**, we got the recommended product for every feature (All, Price, Design, Function) for IC Dictionary, MP3 and Windows NotePC. The results are shown in Table 4.

Table 4: A Comparison of Recommendations

Product Type	Feature	Ranking (high -> low)	System Recommendation
IC Dictionary	All	T2, T5, T4,T1,T3	T5
	Price	T5,T1,T2,T4,T3	T1
	Design	T1,T2,T4,T5,T3	T2
	Function	T2, T5, T4,T3,T1	T5
MP3	All	T7,T9,T8,T6,T10	T7
	Price	T8,T7,T9,T10,T6	T7
	Design	T9,T7,T6,T8,T10	T7
	Function	T7,T9,T8,T6,T10	T7
Windows NotePC	All	T15,T14,T13,T11,T12	T12
	Price	T11,T13,T12,T15,T14	T11
	Design	T14,T15,T11,T13,T12	T14
	Function	T15,T13,T14,T11,T12	T12
SUM	12	4,6,0,0,2	

[Ranking] is the values from User Review from Kakaku.com, sorted in descending order. [System Recommendation] is the recommendation result from our system. The system recommendation agreed with the top suggestion from the User Reviews four times and the second best recommendation six times.

5. Conclusions

This study proposed the entire structure of a decision support system based on the product opinions in blog. In the evaluation of the Recommendation procedure in **4.4**, the validity of the system was confirmed. User Review Graphs from Kakaku.com just provide the review of popular product and can not compare the blog information. Therefore, though some parts of the system need to be improved, the system's construction methods should be recommended.

In the future, we can increase the precision of the whole system through improving the Blog analysis and classifica-

tion method. In the process of Blog analysis, we should study how to distinguish between opinion phrases and general words. Furthermore, we will work on analyzing the negative expressions and adjusting the threshold, to improve the P/N classification. And we also want to extend the recommendation part to a natural conversation with user.

6. References

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7. Appendix

7.1. Product Types List for Test

	Product Type
IC Dictionary	T1: SONY EBR-S7MS
	T2: CASIO XD-ST6200
	T3: SEIKO SR-E8500
	T4: SEIKO SR-E9000
	T5: SHARP Papyrus PW-A8400
MP3	T6: APPLE iPod MA002J/A
	T7: APPLE iPod nano MA004J/A
	T8: PANASONIC D-snap Audio SV-SD750V-A
	T9: SONY NW-A608
	T10: SONY NW-A1000
Windows NotePC	T11: FUJITSU FMV-BIBLO LOOX T70R/T
	T12: NEC LaVie L LL750/ED
	T13: PANASONIC CF-W4HW8AXR
	T14: SONY VAIO VGN-FJ10B
	T15: TOSHIBA dynabook TX/760LS PATX760LS

7.2. User Reviews from Kakaku.com

T1: SONY EBR-S7MS



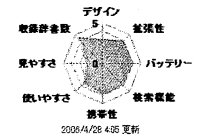
T2: CASIO XD-ST6200



T3: SEIKO SR-E8500



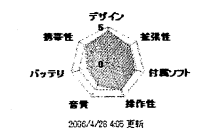
T4: SEIKO SR-E9000



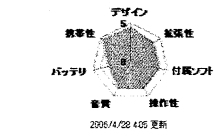
T5: SHARP Papyrus PW-A8400



T6 :APPLE iPod MA002 J/A



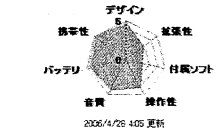
T7: APPLE iPod nano MA004 J/A



T8: PANASONIC SD750V-A



T9: SONY NW-A608



T10: SONY NW-A1000



T11: FUJITSU FMV LOOX T70RT



T12: NEC LaVie L LL750



T13 PANASONIC CF-W4HW8AXR



T14: SONY VAIOVGN-FJ10B



T15: TOSHIBA dynabookTX760LS

